🔷 Section 16 — Unified System Formulation

16.0 System Overview (Final —Version)

The SEKHEM-TSDD (Triangulated Stochastic Decision Dynamics) framework defines a unified decision–dynamical system integrating stochastic processes, probabilistic evolution, network interactions, and control mechanisms within a single closed-loop structure.

The system state is defined as:

$$\Psi_k \in \Omega \subset \mathbb{R}^d, \quad \Omega \text{ is a compact domain}$$

where Ψ_k denotes the system state at discrete time step k, and Ω is a bounded and closed subset of $\mathbb{R}^d$.

The framework incorporates the following key components:

- stochastic dynamics (modeled via stochastic differential equations)

- probability evolution (governed by partial differential equations)
- PAC-based triadic decision mechanisms
- network coupling dynamics
- optimal control formulation

All components interact within a unified mathematical structure, enabling consistent modeling of decision-driven systems under uncertainty.

This formulation establishes the foundation for the unified master equation presented in the next subsection.

SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
DOI: 10.5281/zenodo.19521219"